

*ambarella.com*
3103 Jay Street
Santa Clara, CA 95054
USA

Tel.: +1 408 734 8888

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1.3 ( 08 Feb 2026 )

# ISO 26262 Safety Case

Project:

**CV2A**

Customer:

**Ambarella**

<City, Country>

Contract No.: <CUS xx/xx-xxx>
Report No.: <CUS xx/xx-xxx> <Rxxx>
Version V1, Revision R3, March 2026

Asher Shih

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## Management Summary

The document shows the proposed set-up of an ISO 26262 compliant safety case with requirements, claims, arguments and evidence documents. Furthermore, it shows product-related requirement classes, which *Ambarella* recommends for the development of an automotive functional safety system.

*Ambarella* uses the Safety Case structure together with the tracking to all relevant product safety requirements and ISO 26262 requirements as the argument for completeness. **The Safety Case structure should be agreed with the technical domain experts of the customer, and with the assessor. *Ambarella* makes no absolute claim of completeness and consistency.**

*Ambarella* uses the requirement classes as an argument for completeness and inner consistency of the requirements specification. The requirements classes are, however, only one argument. Another argument is the round-trip requirements engineering process defined by *Ambarella*.

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**Ambarella CV2A**

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# 1 Purpose and Scope of this Document

This document summarizes the safety case according to the ISO 26262:2018 standard for

**Ambarella CV2A**

The purpose of a safety case is to provide a clear, comprehensive and defensible argument, supported by evidence, that functional safety is achieved for the product and that it is free from unreasonable risk when operated in the intended context.

This document covers the following work products from ISO 26262-2:2018, clause 6:

– Safety Case

## 1.1 Reference Documents

This document is based on the *exida* template “ISO 26262 Safety Case”; all rights reserved.

The services delivered by *exida* were performed based on the following norms and standards:

| ID | Document name and revision | Description |
|---|---|---|
| R1 | ISO 26262:2018 (2. Edition) | Road vehicles – Functional safety, Parts 1-12 |
| | | |

The following independent references were used as part of the services delivered by *exida*:

| ID | Document name and revision | Description |
|---|---|---|
| R41 | “A Systematic Approach to Safety Case Management”, Tim Kelly, 2003 | Publication describing the usage of Goal Structured Notation (GSN) in a Safety Case http://www-users.cs.york.ac.uk/~tpk/04AE-149.pdf |
| | | |

## 1.2 Revision History

| Rev. | Date | Status | Author | Notes |
|---|---|---|---|---|
| V0, R1 | Jun 2020 | Draft | Asher Shih | Initial draft based on the exida’s template |
| V0, R2 | Aug 2021 | Draft | Asher Shih | Revised based on exida’s template (V1, R1) |
| V0, R3 | Oct 2021 | Draft | Asher Shih | Cleaned up findings based on the reviewer’s feedback.<br>Added arguments for FuSa audit and assessment.<br>Fixed the table of terms and abbreviations. |
| V0, R4 | Oct 2021 | Draft | Asher Shih | Cleaned up more findings based on the reviewer’s feedback.<br>Updated the revision number of evidence documents. |
| V0, R5 | Nov 2021 | Draft | Asher Shih | Cleaned up the findings identified in the assessment meeting. |
| V0, R6 | Feb 2022 | Release | Asher Shih | Updated the revision number of evidence documents. |
| V1, R1 | Feb 2022 | Release | Asher Shih | CR21649: Updated Figure 4. |

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Reviewed by Hang on 1-Feb-2022

| Rev. | Date | Status | Author | Notes |
|---|---|---|---|---|
| V1, R2 | Jul 2022 | Release | Asher Shih | Updated HSI revision ID for CR#21783, #21885 and #22023. |
| V1, R3 | Nov 2023 | Release | Asher Shih | Updated revision ID of evidence documents for CR#22253. |

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# 2 Introduction to the Safety Case

The safety case should provide a clear, comprehensive and defensible argument, supported by evidence, that functional safety is achieved for a product and that it is free from unreasonable risk when operated in the intended context. The safety case combines three principal elements:

- Safety Objectives
- Safety Arguments
- Safety Evidence

Safety objectives of a product are its top-level safety requirements, and the derived and related safety requirements. Functional safety is achieved if all safety requirements / safety objectives are met. The safety arguments explain how the safety evidence (e.g. ISO 26262 work products) relates to and helps meeting the safety objectives. Both argument and evidence are crucial elements of the safety case that must go hand-in-hand. An argument without supporting evidence is unfounded, and therefore unconvincing. Evidence without an argument is unexplained, resulting in a lack of clarity as to how the safety objectives have been satisfied.

An ISO 26262 safety case combines product arguments (based on technical features or solutions implemented in the product) with process arguments (based on ISO 26262 requirements or another adopted development procedure). Both types of argument should be used to achieve a sound argument for the safety of the product, where a process argument can be seen as providing the confidence in the evidences used in the product argument.

The safety case in this document is constructed to achieve the above goals as following:

Chapter 3 provides an overview of the product, including the intended safety related use cases.

Chapter 4 describes the product's top-level safety requirements, which form the safety objectives for the product. It is the purpose of this safety case to demonstrate, that these safety requirements are met when the product is operated in the intended use cases.

Chapter 5 provides an overview of the product's safety concept, which is one of the main product arguments within the safety case. Individual aspects of the safety concept are picked up again and elaborated further in chapter 7.

Chapter 6 provides an overview of the functional safety management organization during product development, which is an important part of the safety case's process argument.

Chapter 7 contains the main part of the safety case, using a goal structured notation (GSN) to provide a hierarchically structured argument with evidence that the safety objectives are fulfilled.

As shown in Figure 1 below, the GSN uses a tree-like graphical representation to link the individual elements of a safety case (goals, claims and solutions) and to show their relationships:

- A goal is successively broken down into claims
- The achievement of a claim is supported by one or several solutions
- A solution combines the argument and evidence for the achievement of a claim

Figure 2 shows the template, how a goal, claim or solution is described in detail in the safety case throughout chapter 7.

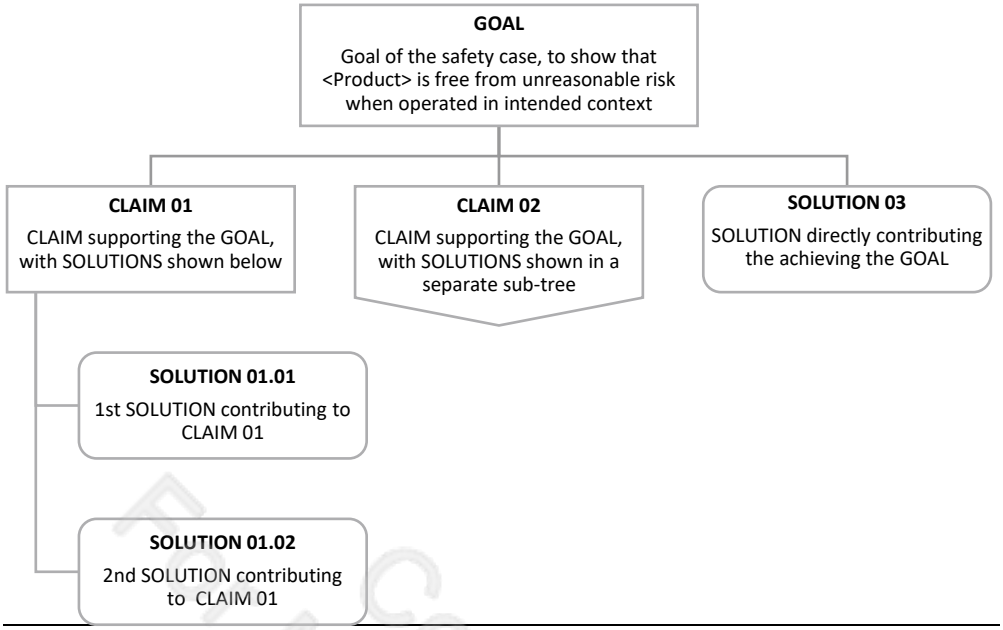



**Figure 1 – Example GSN as used in this safety case**

| **SC-## CLAIM \| SOLUTION: <DESCRIPTION>** | | |
|---|---|---|
| Owner: Person responsible for this part of the safety case | | |
| **ARGUMENT** | | |
| 1. Objective:<br>Description of the objective(s) for the claim, which shall be fulfilled by the individual arguments<br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>Argumentation how the objectives / claim are met, as backed by the evidence work products<br>3. Gaps & Action items:<br>Tracking of open issues. The final safety case shall not have any gaps or open action items.<br>4. Conclusion:<br>Short summary of the status (not started \| in work \| complete \| …) and conclusion | | |
| **EVIDENCE DOCUMENTS** | | |
| <Doc.ID> | <Document name and revision> | <Description> |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-x, Clause y | ISO 26262 clause or work product that is covered by this argument | |
| ***GUIDENCE*** | | |
| *Template text that can be used as a starting point / hint to provide arguments* | | |

**Figure 2 – Template for the detailed description of a GSN goal, claim or solution**



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# 3 Product Overview

## 3.1 Product Description

The CV2A chip is a system on a chip (SoC) for automotive markets, including e-Mirror and ADAS applications, and is targeted to comply with ASIL-C according to ISO 26262. CV2A chip, as known as CV2FS / CV22FS, contains a quad core Arm® Cortex®-A53 for application processing and embedded accelerators for image processing, video encoding, and computer vision/AI functions—especially neural network processing and optical flow. The accelerators are co-processors to the Arm cores. A dual core Cortex-R52 in lockstep is added for safety critical applications and safety monitoring. All processors and co-processors connect through a single controller to access DRAM.

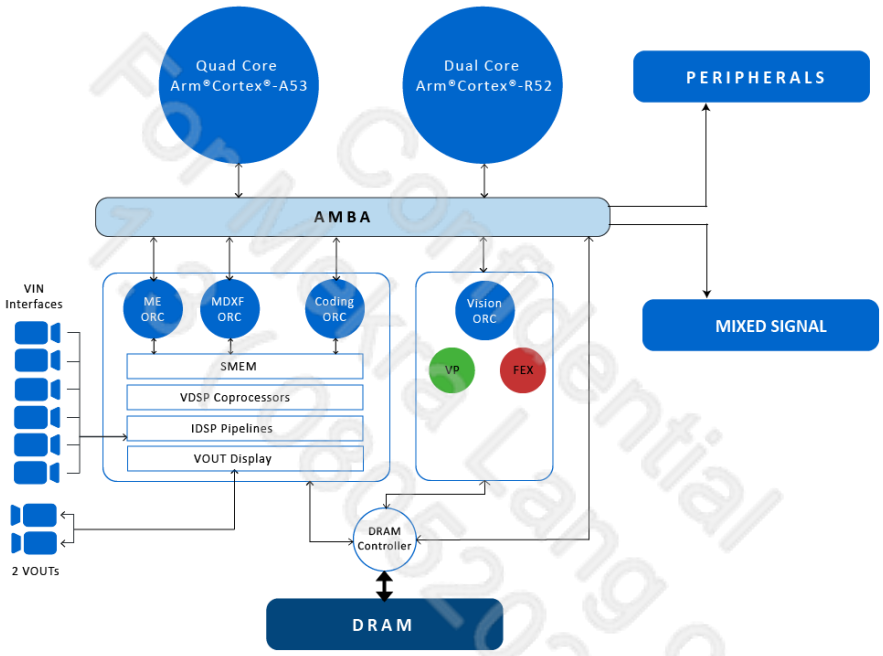



**Figure 3 – Conceptual View of the Architecture**

A more detailed description of CV2A can be found in the SEooC Definition **Error! Reference source not found.**, Safety Manual **Error! Reference source not found.** and Data Sheet [DTS_1000006].

## 3.2 Use Case Description

The following use cases give an overview of the safety related systems and applications, which CV2A is expected to be integrated and used in. The list shown below may not be exhaustive or complete, but these are the typical and most important safety related use cases.

### 3.2.1 Use Case UC-01: Forward-Facing ADAS Camera

Typically, this use case provides a single sensor-based solution, but can be dual or triple sensor-based in certain high-end variants. In this use case, different sensor configurations must be

supported. Additionally, if video streaming or recording functionality is required, CV2A can be configured to output compressed H.264 video as well as object data.

### 3.2.2 Use Case UC-02: Electronic Mirror System

Electronic mirror systems are also known as camera monitor systems (CMS) as defined under ISO 16505. Basic configuration includes in-cabin rearview mirror replacement, which provides a video viewing solution. Potential future CV functionality includes blind spot detection (BSD); CV2A is targeted to support two- and three-channel CMS architectures.

### 3.2.3 Use Case UC-03: Driver State and In-Cabin Monitoring Systems

This use case can include a single camera or dual camera configuration, typically global shutter based for driver state monitoring (DSM); RGB-IR rolling shutter HDR sensors are considered to enable in-cabin monitoring solution. The application monitors a driver's attention by detecting head position, gaze of eyes, as well as eye lid frequency to determine drowsiness. To ensure that variable lighting conditions including drivers wearing sunglasses do not affect the functionality of the application, the system deploys pulsed LED light sources synchronized with the capturing sequence of the image sensor. This use case also considers the addition of RGB-IR sensors for passengers and the monitoring of "lost objects." In this case, near-infrared LEDs are used for consistent lighting.

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# 4 Top-Level Safety Requirements

CV2A has been developed as a Safety Element out of Context (SEooC), based on assumed safety requirements. Table 1 below lists all assumed top-level safety requirements (TLSR) for the SEooC. It is the purpose of this safety case, to provide evidence and arguments that the product fulfills these safety requirements and thereby achieves functional safety.

**Table 1 – Assumed top-level safety requirements for CV2A**

| Req.ID | Requirements Text | ASIL |
|---|---|---|
| **TLSR-1** | The SEooC shall safely acquire image data from an external sensor and prepare captured data for processing by the computer vision system. | C |
| **TLSR-2** | The SEooC shall safely encode image sequences captured from each external sensor according to H.264 standard. | C |
| **TLSR-3** | The SEooC shall safely execute vector operations according to the instructions and algorithms given by software. | C |
| **TLSR-4** | The SEooC shall perform safe general purpose data processing, including:<br>- Correct execution of the defined instruction set<br>- Correct handling of exceptions and interrupts | C |
| **TLSR-5** | For safety-related data, the SEooC shall perform on-chip data transfer and data storage between local memories without loss or corruption safely. | C |
| **TLSR-6** | The SEooC shall safely configure and control the sub-elements. | C |
| **TLSR-7** | The SEooC shall support safe I/O communication with the safety-related external components, including:<br>- NAND, NORSPI, or eMMC device.<br>- CAN, SPI, I2C interface<br>- DRAM (DDR4 / LPDDR4 / LPDDR4x) interface.<br>- sensor(s) via MIPI interface<br>- video output via MIPI-DSI<br>- GPIO interface | C |

For more detailed information about the top-level safety requirements, please refer to the SEooC definition document **Error! Reference source not found.**.

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# 5 Safety Concept Overview

Figure 4 provides and overview of the technical safety concept for CV2A, showing the major safety mechanisms that are implemented in hardware, software or that are defined as assumptions-of-use (AoU) to detect and control internal errors. It also indicates the ASIL allocated to the different IP hardware blocks.

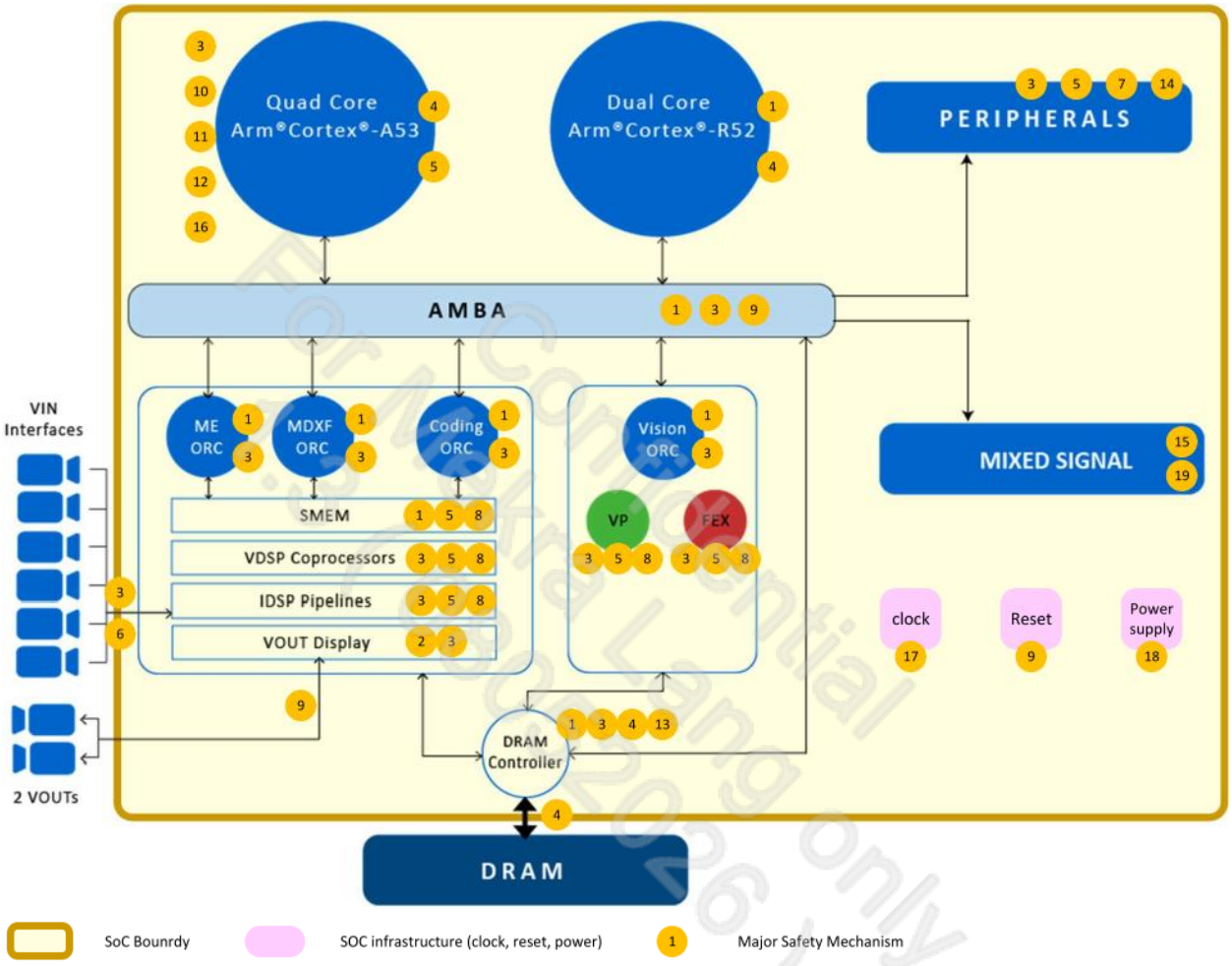



**Figure 4 – SEooC with safety mechanisms and ASIL allocation**

The following safety mechanisms are shown in Figure 4:

①.Dual-core lockstep (DCLS)
②.Duplicated logic (Hardware redundancy)
③.Watchdog timer (or timeout mechanism)
④.ECC
⑤.Parity
⑥.CRC or signature
⑦.End-to-End (E2E) protection protocol
⑧.Software Test pattern
⑨.Software Register check

⑩. Software Data Integrity check
⑪. Software GIC checker (Interrupt monitor)
⑫. Software Test Library (STL)
⑬. Segmentation (Access protection)
⑭. Hardware loop-back
⑮. Built-in self-test pattern (BIST)
⑯. Performance monitor
⑰. Clock monitor
⑱. Power monitor (either internal or external)
⑲. Voltage detector

*Note: Not all safety mechanisms are shown in Figure 4. Only major safety mechanisms, which provide the most significant diagnostic coverage contribution, are included, while smaller safety mechanisms are omitted for the sake of simplicity and to avoid overloading the picture. A detailed description of all safety mechanisms is provided in the Technical Safety Concept* ***Error! Reference source not found.*** *and Safety Manual* ***Error! Reference source not found.****.*

CV2A contains a central error handling unit (CEHU), to consolidate all internal error signals, error interrupts and other error indicators from the hardware modules in one central location and to map these to one or several configurable error reactions. An error reaction can be any of the defined top-level safe states (TLSS). In addition, actions for internal error handling are also available as possible error reactions. These include triggering an interrupt request to a CPU core (i.e. to handle the error via an interrupt service routine), or issuing a programmable software reset (e.g. to mitigate the effect of a transient fault).

Besides consolidating hardware errors, errors detected by software can also be reported to CEHU by setting one of several dedicated SW error indication registers. This triggers the configured error reaction(s) in the same way as for hardware errors.

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# 6 Functional Safety Management

An effective functional safety management organization within the project team and throughout the company, with well-defined roles and responsibilities, aids the ISO 26262 compliant development of a safety related system or element. The following provide an overview of the Ambarella functional safety management organization.

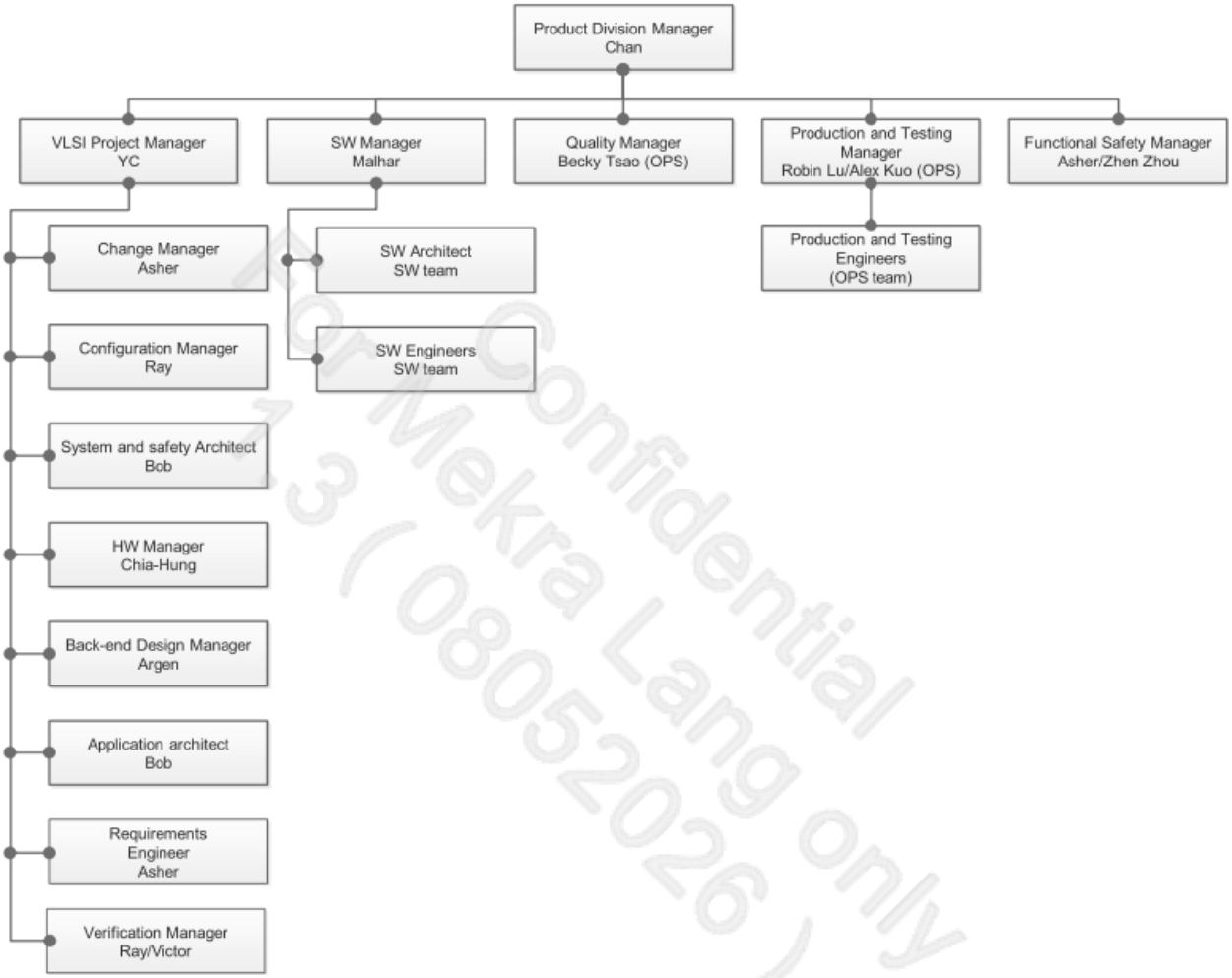



**Figure 5 – Org-chart of Ambarella functional safety management organization**

The most important organizational units and roles are shown below, together with their roles and responsibilities in the CV2A project. For more detailed information, please refer to section 4.3 of the Safety Plan.

**Table 2 – Key roles in Ambarella functional safety management organization**

| Role in CV2A | Organization | Responsibility in CV2A |
|---|---|---|
| Product Manager | VLSI | - Coordinates the top-level product requirements between the stakeholders and the product development team.<br>- Coordinates the project planning with the involved organizational units as shown in the project organization chart.<br>- Coordinates technical issues among the development disciplines and internal organizations and externally, with customer and suppliers.<br>- Obtains and allocates resources to ensure effective implementation of the Plan.<br>- Evaluates the competence of the allocated human resources versus the required competence and ensure necessary enhancement by training and coaching in methods, tools, and techniques used in functional safety tasks. |
| Project Manager | VLSI | - Responsible for the synchronization of the safety plan with the overall project plan.<br>- Acts as the interface to the rest of the project groups through the Product Manager.<br>- Reports any system requirements clarification or misunderstanding to the Product Manager for resolution and clarification.<br>- Coordinates the planning of the technical activities with the involved organizational units as shown in the project organization chart. |
| Functional Safety Manager | VLSI | - Prepares and maintains the Safety Plan.<br>- Monitors and documents the progress of the tasks defined in the Safety Plan.<br>- Coordinates functional safety tasks within the overall context of the system safety program.<br>- Defines and documents the safety architecture.<br>- Coordinates together with Safety Architect and HW project manager technical issues related to functional safety among the development disciplines and internal organizations and externally, with customer and suppliers.<br>- Ensure that adequate records are created to document evidence on performed functional safety activities.<br>- Participates in audits of safety plan implementation.<br>- Performs confirmation measures regarding functional safety activities (Audit, review and assessment of functional safety). |
| SW Manager | SW | - Responsible to manage SW development including SW-related functional safety activities. |

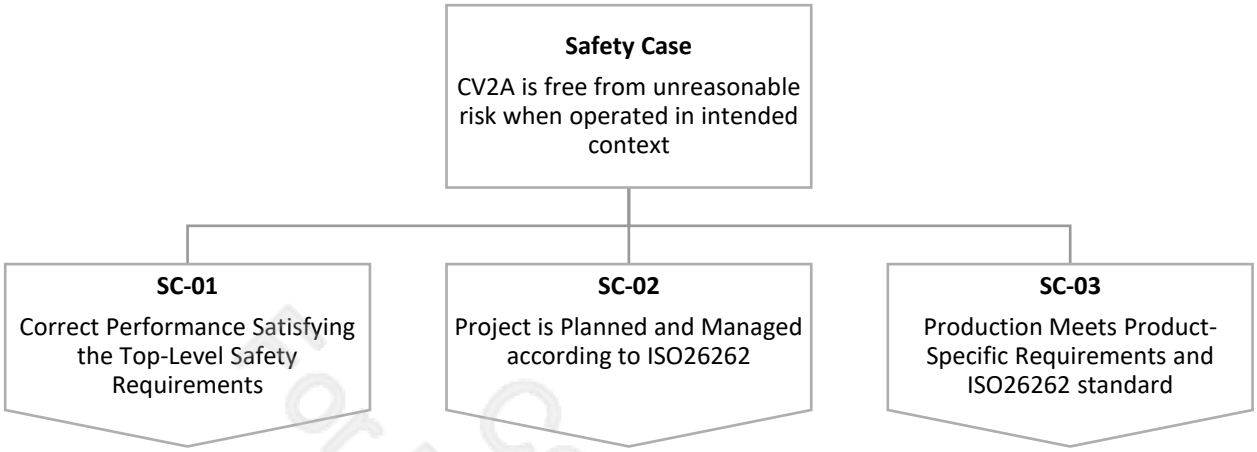
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| Role in CV2A | Organization | Responsibility in CV2A |
|---|---|---|
| Quality Manager | OPS | - Responsible to perform quality tasks within the safety related project. |
| Production and Testing Manager | OPS | - Responsible to manage production and testing. |

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# 7 Safety Case

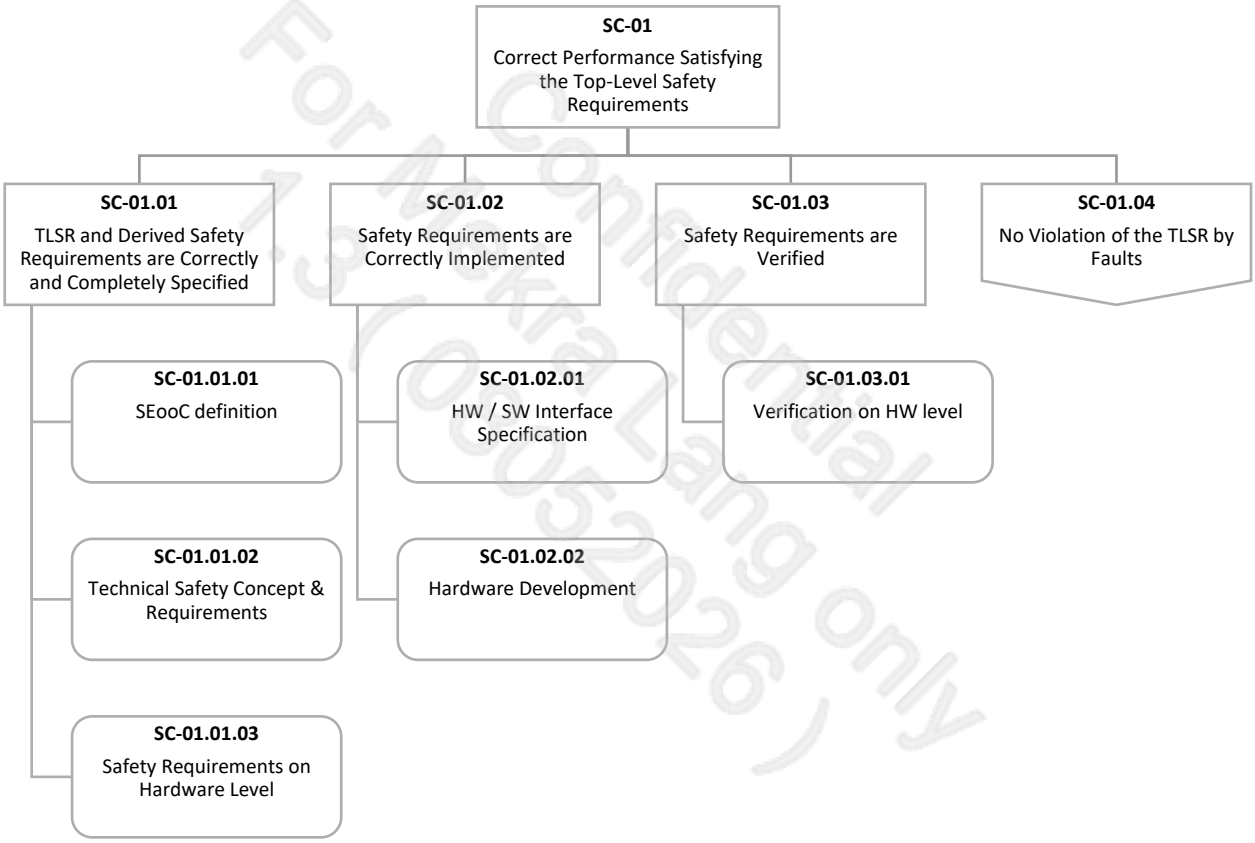
The following shows the overall structure of the safety case, and the top-level claims and solutions to achieve the —




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## 7.1 Correct Performance Satisfying the TLSR (SC-01)

The goal is hierarchically subdivided into sub-claims and solutions with arguments. The following set of claims explains why Ambarella is of the opinion that CV2A meets the applicable objectives and requirements.




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### 7.1.1 TLSR and Derived Safety Requirements are Correctly and Completely Specified (SC-01.01)

The following arguments explain the Ambarella requirements capturing, refinement and tracking process (Requirements Engineering) and how the process meets the applicable ISO 26262 requirements from parts 3, 4, 5, 6 and 8.

| **SC-01.01 CLAIM: TLSR and Derived Safety Requirements are Correctly and Completely Specified** |
|---|
| Owner: Asher Shih, Yu-Ching Su |
| **ARGUMENT** |
| 1. Objective:<br>1.1 Provide arguments and evidence that the specification of the safety requirements are complete, correct and consistent.<br>1.2 Provide arguments and evidence that the assumptions on the SEooC or System environment and safety-related use are sufficient and communicated to the User or System Integrator.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>2.1 Argument on Procedures and Workflow:<br>The procedure and workflow for defining and refining requirements at Ambarella is defined in the requirements engineering process. Requirements are put under configuration management by using Polarion.<br>See also<br>- SC-01.01.02: Technical Safety Concept & Requirements<br>- SC-01.01.03: Safety Requirements on Hardware Level<br><br>2.2 Arguments for the Completeness, Correctness, and Consistency of the TLSRs:<br>The claim is achieved by the Ambarella requirements engineering process supported by requirements management tool. See Ambarella process specification [D42963_0].<br>Ambarella derived technical safety requirements.<br><br>2.3 Arguments for the Completeness, Correctness, and Consistency of the Derived Safety Requirements:<br>Completeness and correctness of the requirements that were input to the project is ensured by reviews. Based on these top-level requirements, the derived requirements were reviewed for consistency (inspection method for ASIL-C). The requirements were completed by performing analyses: FMEA on System-level, FMEDA on HW-Level and DFA on System-level.<br>See also:<br>- SC-01.01.03: Safety Requirements on Hardware level<br><br>2.4 Arguments for the Adequacy of the Assumptions:<br>The assumptions on proper installation of the CV2A and environmental conditions the CV2A can be exposed to, are referenced in the safety manual. |

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3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for safety requirements completeness and correctness are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D40599_0 | CV2A_Safety_Case, V0-R5 | Safety Case |
| D42963_0 | Ambarella_Safety_Requirement_Development_Flow.docx, v0.1 | Safety Requirements Specification methodology |
| VLSI-1545 | Ambarella CV2FS CV22FS HW Safety Manual, v0.6 | Safety Manual |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-8, Clause 6 | Specification and management of safety requirements | |

## SC-01.01.01 SOLUTION: SEooC Definition

Owner: Asher Shih, Yu-Ching Su

**ARGUMENT**

1. Objective:

The objectives of the SEooC Definition are:

a) to define and describe the Safety Element out of Context (SEooC), its functionality, assumed allocated requirements, dependencies on, and interaction with the environment and other elements in the system
b) to support an adequate understanding of the SEooC so that the activities in subsequent phases can be performed

2. Argument for meeting the objective and related ISO 26262 requirements:

During the development of the SEooC, use cases are first defined based on the product requirements. The top-level requirements, including functional requirements and safety requirements, are specified for the activities in subsequent phases of development. The top-level safe states, boundaries and constraints are also specified for the integration of the SEooC into the system. The completeness and correctness of the SEooC definition is ensured by inspection review based on the *exida* template, “ISO 26262:2018 SEooC Definition – Review Report”.

3. Gaps & Action items:

None.

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| | | |
|---|---|---|
| 4. Conclusion:<br>The argument shows that the objectives of the SEooC Definition are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| VLSI-1456 | ISO 26262 Ambarella Safety Element out of Context, rev. 1.03 | The SEooC definition |
| D35680_0 | CV2A_SEooC_Definition_Review_Report, v1.4 | Review report of the SEooC definition |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO26262-10, Clause 9 | Safety Element out of Context | |

**SC-01.01.02 SOLUTION: Technical Safety Concept & Requirements**

Owner: Chia-Hung Cheng

**ARGUMENT**

1. Objective:

The objectives for the Technical Safety Concept (TSC) and Technical Safety Requirements (TSR) are:

a) to specify technical safety requirements regarding the functionality, dependencies, constraints and properties of the SEooC elements and interfaces needed for their implementation
b) to specify technical safety requirements regarding the safety mechanisms to be implemented in the SEooCelements and interfaces
c) to specify requirements regarding the functional safety of the SEooCand its elements during production, operation, service and decommissioning
d) to verify that the technical safety requirements are suitable to achieve functional safety at the system level and are consistent with the functional safety requirements
e) to develop a chip-level architectural design and a technical safety concept that satisfy the safety requirements and that are not in conflict with the non-safety-related requirements
f) to analyze the chip-level architectural design in order to prevent faults and to derive the necessary safety-related special characteristics for production and service
g) to verify that the chip-level architectural design and the technical safety concept are suitable to satisfy the safety requirements according to their respective ASIL

2. Argument for meeting the objective and related ISO 26262 requirements:

2.1 Arguments on TSC:

Ambarella's TSC is developed based on the *exida* template, "ISO 26262 Technical Safety Concept for Semiconductor IC and IP", to satisfy the technical safety requirements in the context of the defined SEooC design.

In detail the technical safety concept is developed where the following properties apply:

- technical safety requirements are allocated to the SEooC design components (illustrated in the sections of "System Architectural Design and TSR Allocation").

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- the ability to execute tests during system integration is given (in the chapter of Safe State Concept or in the section of “Requirements for integration and operation”),
- all internal and external interfaces of safety-related components are defined,
- modularity, consistent level of granularity and simplicity are taken care of, and
- if possible, well trusted design principles are used

2.2 Arguments on the Correctness, Completeness and Consistency of the TSR:

Technical Safety requirements specification is existing, which compiles

- the Technical Safety requirements derived from the Top-Level Safety requirements
- the Technical Safety requirements to meet the specified safety timings (in DTTI field)
- the Safety Mechanism requirements to meet the specified ASIL.

The Technical Safety requirements specification has a pre-defined hierarchical structure shown in section 2.3 of Technical Safety Concept.

The Technical Safety requirements are linked to the chip-level design architecture and Technical Safety Concept (illustrated in the sections of “System Architectural Design and TSR Allocation”).

The following properties are addressed by the technical safety requirements specification:

- the external interfaces, such as communication and user interfaces, if applicable;
- the constraints, e.g. environmental conditions or functional constraints; and
- the system configuration requirements; and
- the response of the system or elements to stimuli that affect the achievement of safety goals; and
- the necessary safety mechanisms including detection, indication and control of faults within the system and external to the system; and
- the necessary safety mechanisms to achieve or maintain a safe state including (1) the details on the transition to the safe state, (2) diagnostic test time interval, (3) the measures to maintain the safe state; and
- the necessary safety mechanisms to prevent faults from being latent.

See also

- SC-01.03: Safety Requirements are verified

3. Gaps & Action items:

None.

4. Conclusion:

The argument shows that the objectives for the Technical Safety Concept and Technical Safety Requirements are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D11989_0 | CV2A_Technical_Safety_Concept, V1-R4 | Technical Safety Concept |
| D18783_0 | CV2A_Requirements, r605491. | Technical Safety Requirements Specification |
| D36989_0 | CV2A_TSC_Review_Report_Asher, V1-R4 | TSC / TSR Verification Report |

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| | | |
|---|---|---|
| D39692_0 | CV2A_TSC_Review_Report_Karthik, V1-R4 | |
| D36989_1 | CV2A_requirement_traceability_review, V1-R4 | |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-4, Clause 6<br>ISO 26262-9, Clause 5 | Technical safety concept<br>Requirements decomposition with respect to ASIL tailoring | |

| **SC-01.01.03 SOLUTION: Safety Requirements on Hardware Level** | | |
|---|---|---|
| Owner: Chia-Hung Cheng | | |
| **ARGUMENT** | | |
| 1. Objective:<br>The Hardware Safety Requirements (HWSR) are derived from the Technical Safety Concept, System Architectural Design Specification, and the HW/SW Interface Specification, and they shall provide sufficiently detailed information to enable the design, implementation and verification of the HW.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br><br>2.1 Arguments on the Correctness, Completeness and Consistency of the HWSR:<br>For IC development, the Hardware Safety Requirements (HWSR) are equivalent to the technical safety requirements (TSR) allocated in hardware. A lower level HWSR is developed to refine TSR with explicit description so that the HW designer is able to follow up the implementation. The correctness, completeness and consistency is reviewed by inspection based on the *exida* template, "ISO 26262:2018 Technical Safety Concept & Requirement Review report".<br>See also<br>- SC-01.03: Safety Requirements are verified<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The argument shows that the objective for Safety Requirements on Hardware Level are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D18783_0 | CV2A_Requirements, r605491 | Hardware (Safety) Requirements Specification |
| D36989_0 | CV2A_TSC_Review_Report_Asher, V1-R4 | Hardware Requirements Verification Report |

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| | | |
|---|---|---|
| D39692_0 | CV2A_TSC_Review_Report_Karthik, V1-R4 | |
| D36989_1 | CV2A_requirement_traceability_review, V1-R4 | |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-5, Clause 6 | Specification of hardware safety requirements | |

### 7.1.2 Safety Requirements are Correctly Implemented (SC-01.02)

The following arguments explain that the Ambarella requirements are correctly implemented and derived into the design domains.

| **SC-01.02 CLAIM: Safety Requirements are Correctly Implemented** |
|---|
| Owner: HW designers |
| **ARGUMENT** |
| 1. Objective:<br>1.1 Objective is that design & implementation of the TLSR and safety functions specified by safety requirements associated with the specified use and environmental conditions of the element are correct and consistent.<br>1.2 Another Objective is that systematic single-point faults do not violate the safety requirements or are mitigated, or do not violate beyond an unacceptable probabilistic acceptable level specified by the HW architectural metrics.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>See<br>- SC-01.02.01: HW / SW Interface Specification<br>- SC-01.02.02: Hardware Development<br><br>The safety requirements specified in the technical safety requirement which was reviewed by Ambarella. They were completely implemented in the CV2A. The requirement traceability is used to check and confirm the completeness of the implementation of safety requirements.<br>The hardware design was reviewed and analysed by FMEA, FMEDA, and tested by HW Fault insertion tests and functional testing as well. The results of the analyses and tests were reviewed. The correct requirement implementation was verified by functional testing.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for safety requirements are met. |

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| EVIDENCE DOCUMENTS | |
|---|---|
| See evidence documents in SC-01.02.01, and SC-01.02.02. | |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | |
| ISO 26262-5, Clause 5 | Specification of hardware safety requirements |
| ISO 26262-4, 6.4.7<br>ISO 26262-5, 6.5.2 | Hardware-software interface (HSI) specification<br>HSI specification (refined for HW requirements) |
| ISO 26262-5, Clause 7 | Hardware Design |

**SC-01.02.01 SOLUTION: HW / SW Interface Specification**

Owner: Bob Kunz

**ARGUMENT**

1. Objective:

The HW/SW Interface (HSI) specification shall specify the hardware and software interaction and shall be consistent with the technical safety concept.

The HSI specification shall list all hardware devices that are controlled by software and hardware resources that support the execution of software.

2. Argument for meeting the objective and related ISO 26262 requirements:

The HSI spec provides details on the hardware and software interface, and on how to find the register layout for each sub-component in CV2A. The spec is documented in two parts, one in Ambarella CV2FS CV22FS Hardware Software Interface [VLSI-2196] and the other is in CV2FS Hardware Programming Reference Manual [CV2FS-S03-001]. The former one focuses on Cortex units, embedded processors, and hardware co-processors. Guidelines for safety mechanisms implemented in software is also included here. The latter one covers the programming information for all peripherals and storage devices on the AMBA bus.

Both documents forms HSI spec and are reviewed by inspection.

3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for HW/SW Interface Specification are met.

| EVIDENCE DOCUMENTS | | |
|---|---|---|
| VLSI-2196 | Ambarella CV2FS CV22FS Hardware Software Interface, v0.14 | HW/SW Interface (HSI) Specification |
| CV2FS-S03-001 | CV2FS Hardware Programming Reference Manual, v1.0 | |
| - | The files under doc/review/HSI_review_report | HSI Verification Report |

| ISO 26262 CLAUSES AND REQUIREMENTS COVERED | |
|---|---|
| ISO 26262-4, 6.4.7 | Hardware-software interface (HSI) specification |
| ISO 26262-5, 6.5.2 | HSI specification (refined for HW requirements) |

| SC-01.02.02 SOLUTION: Hardware Development | | |
|---|---|---|
| Owner: HW designers | | |
| **ARGUMENT** | | |
| 1. Objective:<br>The hardware design shall comply with the allocated hardware requirements and the safety requirements specification. Compliance of the objective is verified according to methods of ISO 26262 and with analyses like DFMEDA, FMEDA, DFA, etc.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>Arguments for HW Development:<br>- A lot of Hardware design are re-used from the previous generation of chips.<br>- Well trusted design methods are used for HW design and design complexity is controlled.<br>- The TSC and HW micro-architecture specification contains a detailed hardware description<br>- Inductive analysis are used like FMEA, FMEDA, DFA<br>- Requirements traceability shows completeness and consistency between the TSR, hardware safety requirements and their implementation<br>- Operating conditions and influences, of temperature, vibrations, water, EMI, crosstalk originating from its environment are covered within the DFA<br>- Evidence of the effectiveness of safety mechanisms is confirmed by FMEDA and fault insertion testing<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for hardware development are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| - | See documents mapped to WP5-7.5.1 in CV2A_WP_doc_mapping_table.xlsx [D37409_0] | Hardware Design Specification |
| - | See documents mapped to WP5-7.5.3 in CV2A_WP_doc_mapping_table.xlsx [D37409_0] | HW Design Verification Report |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-5, Clause 5 | General topics for the product development at the hardware level | |

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| | |
|---|---|
| ISO 26262-5, Clause 7 | Hardware design |

### 7.1.3 Safety Requirements are Verified (SC-01.03)

The following arguments explain that the Ambarella requirements are correctly verified.

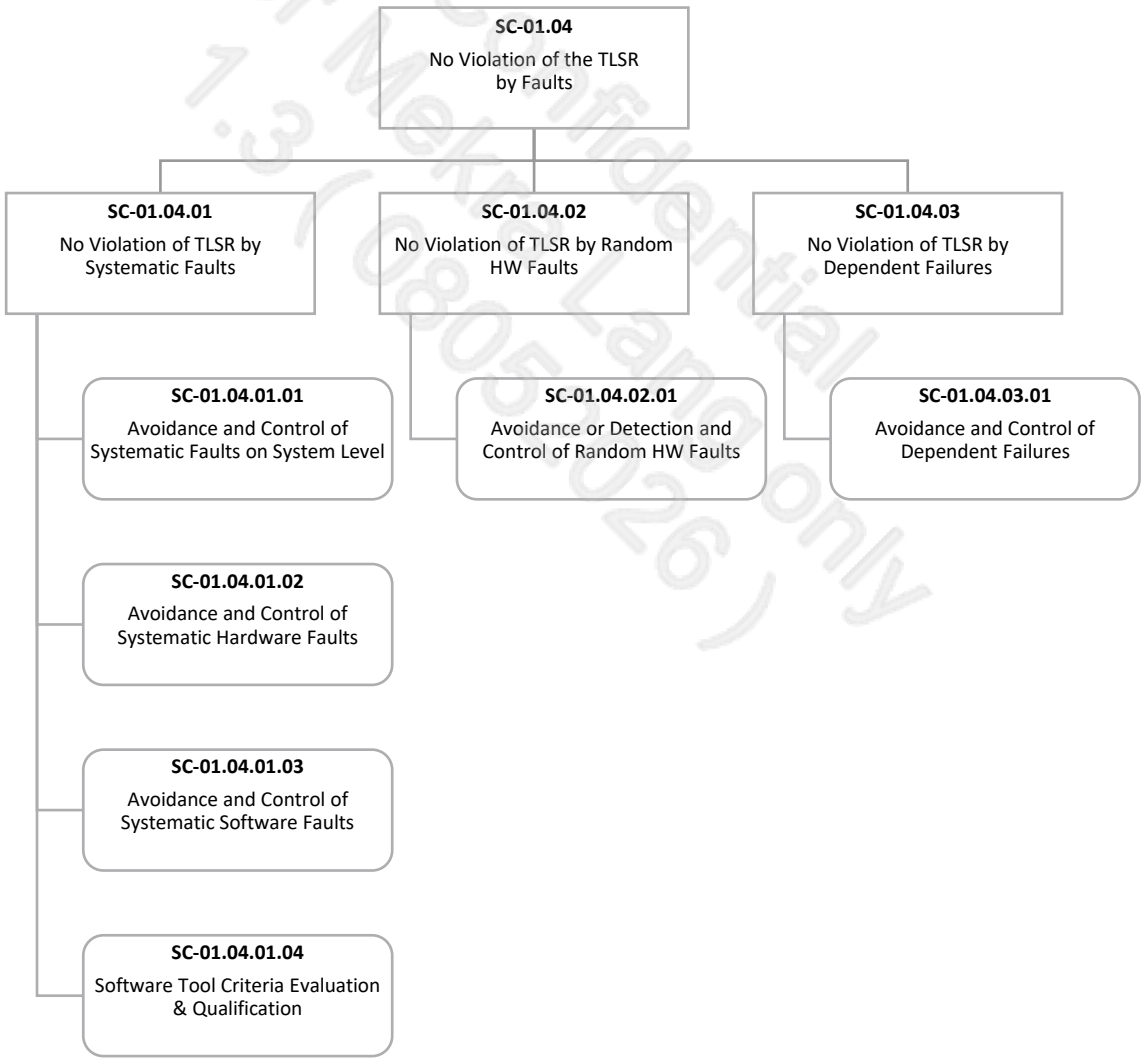
| **SC-01.03.01 SOLUTION: Verification on HW level** | | |
|---|---|---|
| Owner: Verification engineers | | |
| **ARGUMENT** | | |
| 1. Objective:<br>The objective is to ensure the compliance of the developed hardware with the hardware safety requirements.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>The following verification strategy for HW was planned by Ambarella as a combination of reviews, analyses and tests to cover all ISO 26262 test requirements:<br>- Inspection of all HW requirements, incl. HW safety requirements<br>- Inspection of HW architecture and design specs<br>- Safety and dependent failure analysis for all HW elements<br>- Pre-silicon, Post-silicon and Production tests based on the HW verification plan. These include simulation, analyses and other verification methods (such as code coverage review, traceability matrixes) based on HW design specification, as well as ATE, characterization, and qualification on post-silicon phase.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for verification on HW level are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D11978_0 | CV2A_Verification_Plan.docx, V1 R1 | HW Verification Plan |
| D40534_0 | CV2A_Verification_Plan_Status.xlsx, ver.7. | HW Verification Plan Status |
| - | See documents mapped to WP 8-9.5.2 in CV2A_WP_doc_mapping_table.xlsx [D37409_0] | HW Verification Specifications |
| - | See documents mapped to WP 8-9.5.3 in CV2A_WP_doc_mapping_table.xlsx [D37409_0] | HW Verification Report |

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| ISO 26262 CLAUSES AND REQUIREMENTS COVERED | |
|---|---|
| ISO 26262-5, 6.5.3 | Verification of HW safety requirements |
| ISO 26262-5, 7.5.3 | Verification of HW design spec + HW safety analyses |
| ISO26262-5, Clause 10 | Hardware integration and verification |
| ISO 26262-8, Clause 9 | Verification |

### 7.1.4 No Violation of a TLSR in the Presence of a Fault (SC-01.04)

The following arguments explain why Ambarella is of the opinion that CV2A meets the applicable functional safety objectives / requirements for fault / failure avoidance and control.




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1.3 ( 08052026 )

| SC-01.04 CLAIM: No Violation of the TLSR by Faults | |
|---|---|
| Owner: Chia-Hung Cheng, and HW designers | |
| **ARGUMENT** | |
| 1. Objective:<br>The objective is the correct performance of the device per the Top-Level Safety Requirements under fault-free and faulty conditions and under all defined environmental conditions.<br>The term “correct performance” includes the transfer into the defined Safe State if correct performance per the Top-Level Safety Requirements cannot be guaranteed anymore.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>This claim is structured into claims and arguments for Systematic Faults, Random HW Faults, and Common Cause Failures, which should also include other systems or elements:<br>- SC-01.04.01: No Violation of TLSR by Systematic Faults<br>- SC-01.04.01.01: Avoidance and Control of Systematic Faults on System Level<br>- SC-01.04.01.02: Avoidance and Control of Systematic Hardware Faults<br>- SC-01.04.01.03: Avoidance and Control of Systematic Software Faults<br>- SC-01.04.01.04: Software Tool Criteria Evaluation & Qualification<br>- SC-01.04.02: No Violation of TLSR by Random HW Faults<br>- SC-01.04.03: No Violation of TLSR by Dependent Failures<br><br>2.1 Argument for “All Safety Mechanisms Required to Control Systematic, Random and Common Cause Failures are Correctly Specified, Implemented and Verified”:<br>For this argument, refer to:<br>- SC-01.01.02: Technical Safety Concept & Requirements<br>- SC-01.01.03: Safety Requirements on Hardware Level<br>- SC-01.02: Safety Requirements are Correctly Implemented<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for no violation of the TLSR by faults are met. | |
| **EVIDENCE DOCUMENTS** | |
| Refer to the evidence documents in section 7.1.4.1 to 7.1.4.3. | |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | |
| ISO 26262-4, 6.4.4 | Safety Analyses and avoidance of systematic failures |
| ISO 26262-5, 7.4.3 | Hardware Safety Analyses |
| ISO 26262-5, Clause 8 | Evaluation of the hardware architectural metrics |
| ISO 26262-5, Clause 9 | Evaluation of safety goal violations due to random HW failures |
| ISO26262-8, Clause 11 | Confidence in the use of software tools |
| ISO 26262-9, Clause 6 | Criteria for coexistence of elements |
| ISO 26262-9, Clause 7 | Analysis of dependent failures |

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| | |
|---|---|
| ISO 26262-9, Clause 8 | Safety Analyses |

### 7.1.4.1 No violation of TLSR by systematic faults (SC-01.04.01)

| SC-01.04.01 CLAIM: No Violation of TLSR by Systematic Faults | |
|---|---|
| Owner: (Refer to SC-02.01, SC-02.04, SC-01.04.01.01 to SC-01.04.01.03) | |
| **ARGUMENT** | |
| 1. Objective:<br>The objectives shall show that no top-level safety requirement is violated due to systematic faults.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>For systematic faults, typically the focus is on fault avoidance. The avoidance of systematic faults can be split into:<br>- Avoidance by overall Functional Safety Management and through the thorough execution of the Safety Plan<br>- Avoidance by conduction of Confirmation Measures<br>- Avoidance of systematic faults by System, HW and SW safety verification<br>- Safety analysis for systematic faults including system-FMEA, design-FMEA, and DFA are performed.<br>Refer to:<br>- SC-02.01: Safety Planning & Safety Management<br>- SC-02.04: Confirmation Measures<br>- SC-01.04.01.01: Avoidance and Control of Systematic Faults on System Level<br>- SC-01.04.01.02: Avoidance and Control of Systematic Hardware Faults<br>- SC-01.04.01.03: Avoidance and Control of Systematic Software Faults<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for no violation of the TLSR by systematic faults are met. | |
| **EVIDENCE DOCUMENTS** | |
| Refer to the evidence documents of SC-02.01, SC-02.04, SC-01.04.01.01, SC-01.04.01.02, and SC01.04.01.03. | |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | |
| ISO 26262-9, Clause 8 | Safety Analyses |

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| SC-01.04.01.01 SOLUTION: Avoidance and Control of Systematic Faults on System Level | | |
|---|---|---|
| Owner: Bob Kunz | | |
| **ARGUMENT** | | |
| 1. Objective:<br>1.1 Causes of systematic faults in the overall system and their effects must be identified and analyzed in order to avoid their causes or to control their effects.<br>1.2 Verify by means of safety analyses that systematic failures do not lead to a violation of the safety requirements considering the specified safety concept.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>Primary focus on is the avoidance of systematic faults.<br>The inductive analyses (FMEA) was planned in the safety plan and systematically carried out during the development phase in the team. The results were reviewed in the team. The open action and tasks were consistently controlled. The Ambarella development process according to the development process description ensures the avoidance of systematic faults. Formal and informal reviews and inspections ensured the completeness, correctness and consistency of requirements and design.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for avoidance and control of systematic faults on system level are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D17352_0 | CV2A_System_Level_FMEA.xlsx, v1.0 | System FMEA |
| D37854_0 | cv2a_System_FMEA_review_report_jsg.docx, v0.6 | System Safety Analysis Verification Report |
| D37854_1 | cv2a_System_FMEA_review_report_jtsai.docx, v0.5 | |
| D42996_0 | Ambarella_Safety_Analysis_methodology.docx, v0.2 | FMEA Process Description |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-4, 6.4.4<br>ISO 26262-9, Clause 8 | Safety Analyses and avoidance of systematic failures<br>Safety Analyses | |

| SC-01.04.01.02 SOLUTION: Avoidance and Control of Systematic Hardware Faults |
|---|
| Owner: HW designers |

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| ARGUMENT | | |
|---|---|---|
| 1. Objective:<br>1.1 Causes of systematic hardware faults and their effects must be identified and analyzed in order to avoid their causes or to control their effects.<br>1.2 Verify by means of safety analyses that systematic failures do not lead to a violation of the safety requirements considering the specified safety concept.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>2.1 Argument on DFMEA:<br>The deductive FMEA was first performed on system level for the avoidance of systematic faults. The deductive Design FMEA (DFMEA) according to AIAG/VDA standard was planned as improvement to cover system FMEA in the safety plan, and was performed for the critical modules, including<br>- New blocks or blocks with non-trivial changes<br>- Analog mixed-signal blocks, especially if they are not trivial and standard<br>- Blocks where we had problems in the past<br>The analysis result was carried out during the hardware design phase in the team. The results were reviewed by inspection. The open action and tasks were consistently controlled.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for avoidance and control of systematic hardware faults are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| - | Files under doc/safety/DFMEA directory | Design FMEA |
| - | Review reports under doc/review/DFMEA_review_report directory | Hardware Safety Analysis Verification Report |
| D42996_0 | Ambarella_Safety_Analysis_methodology.docx, v0.2 | Design FMEA Process Description |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-5, 7.4.3<br>ISO 26262-9, Clause 8 | Hardware Safety Analyses<br>Safety Analyses | |

| SC-01.04.01.03 SOLUTION: Avoidance and Control of Systematic Software Faults |
|---|
| Owner: Customer |
| **ARGUMENT** |
| 1. Objective: |



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1.1 Causes of systematic SW faults and the effects must be identified and analyzed in order to avoid their causes or to control their effects.

1.2 Verify by means of SW safety analysis that systematic SW failures do not lead to a violation of the safety requirements considering the specified safety concept.

2. Argument for meeting the objective and related ISO 26262 requirements:

CV2A is a hardware component, and hence product development on SW level was tailored out in the safety lifecycle. If the algorithms running on CPU are incorrect, hardware has no way of detecting or preventing this. An assumption-of-use is developed and documented in the safety manual [AR-31] for customers to verify the algorithms perform correctly and within accepted ASIL.

3. Gaps & Action items:

None.

4. Conclusion:

SW development is tailored out in this project.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D18783_0 | CV2A_Requirements, r605491 | Technical Safety Requirements Specification |
| VLSI-1545 | Ambarella CV2FS CV22FS HW Safety Manual, v0.6 | Safety Manual |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-9, Clause 8 | Safety Analyses | |

## SC-01.04.01.04 SOLUTION: Software Tool Criteria Evaluation & Qualification

Owner: Victor Melamed and SW tool users

**ARGUMENT**

1. Objective:

The objectives for SW tool evaluation and qualification are:

a) to provide criteria to determine the required level of confidence in a software tool when applicable
b) to provide means for the qualification of the software tool when applicable, in order to create evidence that the software tool is suitable to be used to support the activities or tasks required by ISO 26262 (i.e. the user can rely on the correct functioning of a software tool for those activities or tasks required by ISO 26262).

2. Argument for meeting the objective and related ISO 26262 requirements:

2.1 Argument on SW tool evaluation:

SW tools used for the project development were listed in tool_evaluation.list.xlsx [D7855_0], and were analyzed with the results documented in SW tool criteria evaluation reports.

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2.2 Argument on SW tool qualification:

Software tool evaluation results in TCL1 for all software tools. No SW tools were classified as TCL2 and TCL3 and hence the qualification of the software tools is not applicable to CV2A.

3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for software tool criteria evaluation and qualification are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| - | Documents under doc/methodology/tool_eval/ | Software Tool Criteria Evaluation Reports |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO26262-8, Clause 11 | Confidence in the use of software tools | |

### 7.1.4.2 No violation of TLSR by random HW faults (SC-01.04.02)

| **SC-01.04.02 CLAIM: No Violation of TLSR by Random HW Faults** |
|---|
| Owner: All HW designers |
| **ARGUMENT** |
| 1. Objective:<br>1.1 Random HW faults shall not lead to violation of TSRs beyond specified probabilistic limits.<br>1.2 Dangerous random HW faults shall not lead to violation of the Top-Level Safety Requirements with a probability of > 10 FIT.<br>1.3 Dangerous HW faults shall be detected with SPFM >= 97%, and LFM >= 80% and lead to the transfer into the defined Safe State.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>In case of random HW faults (RHWF), ISO 26262 puts emphasis on the detection and control of faults. Additional measures are specified to avoid or reduce the probability of random HW faults. The detection and control are ensured by implementing safety mechanisms for which completeness has been ensured by deriving them through safety analysis FMEDA, and DFA.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for no violation of TLSR by random HW faults are met. |

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| EVIDENCE DOCUMENTS | | |
|---|---|---|
| D7855_0 | cv2a_fmeda_pnr.xlsx, v0.7 | FMEDA reports |
| D6854_0 | CV2A_DFA_spreadsheet.xlsx, V1-R3 | DFA report |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-5, Clause 8 | Evaluation of the hardware architectural metrics | |
| ISO 26262-5, Clause 9 | Evaluation of safety goal violations due to random HW failures | |
| ISO 26262-9, Clause 8 | Safety analyses | |

**SC-01.04.02.01 SOLUTION: Avoidance or Detection and Control of Random HW Faults**

Owner: All HW designers

**ARGUMENT**

1. Objective:

Verify by quantitative safety analysis techniques that the required random HW architectural metrics (SPFM, LFM) and probabilistic metric for random HW faults (PMHF) are met with the specified technical safety concept, HW design and safety mechanisms.

2. Argument for meeting the objective and related ISO 26262 requirements:

A quantitative FMEDA was performed to analyze the HW detailed design and to determine the hardware metrics PMHF, SPFM and LFM as planned in the Safety Plan.

The random HW architectural metrics of PMHF, SPFM, LFM were calculated for the Passenger Compartment mission profile, with the junction/ambient temperature difference as documented in the permanent base failure rate calculation report.

The values for the SPFM of 97.01% and LFM of 96.85% are also reported. The residual failure rate of 8.48 FIT is also reported for the SEooC (including permanent faults on silicon and package, and transient faults).

The results show that the metrics for ASIL C are met. Implemented safety mechanisms were tested by HW fault insertion testing. The safety requirements specified within the technical safety concept were fulfilled.

2.1 Argument on the Estimation of Base Failure Rates:

The base failure rates for the quantitative analysis was derived from the following sources:

- Permanent faults in the silicon die: IEC TR 62380:2004, Chapter 7
- Passenger compartment profiles: IEC TR 62380:2004, Table 11
- Permanent faults package: IEC TR 62380:2004, Chapter 7
- Transient faults in SRAM, DFFs, latches: Samsung's Foundry data for 10 nm Low Power Plus (10LPP) process.

To avoid an artificial failure rate reduction, the failure rate calculation is based on operating time only, without consideration for OFF-time. This is achieved by forcing $\tau_{off} = 0$.

2.2 Argument on FMEDA for HW Element Level:

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FMEDA is performed on silicon and package separately. For package FMEDA, please refer to [D35139_8]. For silicon FMEDA, the permanent base failure rate is documented in [D35139_247], and the result is documented in [D35140_0] and its sub-reports.

2.3 Argument on Reporting FMEDA results to customers:

FMEDA results are reported to customers in the safety manual. Detailed information about failure rate and FMEDA metrics results are given for silicon permanent, silicon transient and package faults separately, with further break down to top-level failure modes.

3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for avoidance or detection and control of random hardware faults are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D35139_247 | cv2a_fmeda_pnr.xlsx, v0.7 | FMEDA for silicon |
| D35139_8 | CV2A_FMEDA_package_14x14_441.xlsx, v1.50 | FMEDA for package |
| D35139_247 | cv2a_permanent_base_failure_rate_calc.xlsx, v0.28 | Failure Rate Calculation Process Description |
| VLSI-1545 | Ambarella CV2FS CV22FS HW Safety Manual, v0.6 | Safety Manual |
| - | See documents mapped to WP5-8.5.2 in CV2A_WP_doc_mapping_table.xlsx [D37409_0] | FMEDA review reports |
| D7468_0 | cv2a_presilicon_asil_verification_and_fmeda_generation_methodology.docx, v3.4 | FMEDA Process Description |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-5, 7.4.3 | Safety analyses | |
| ISO 26262-5, Clause 8 | Evaluation of the hardware architectural metrics | |
| ISO 26262-5, Clause 9 | Evaluation of safety goal violations due to random HW failures | |
| ISO 26262-5, Annex C | Hardware architectural metrics | |
| ISO 26262-9, Clause 8 | Safety analyses | |

### 7.1.4.3 No violation of TLSR by dependent failures (SC-01.04.03)

| **SC-01.04.03 CLAIM: No Violation of TLSR by Dependent Failures** |
|---|
| Owner: Alan Zhu, Praveen Jaini, Samy |



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| ARGUMENT | | |
|---|---|---|
| 1. Objective:<br>The objectives are:<br>a) to confirm that a required independence or freedom from interference is sufficiently achieved in the design by analyzing their potential causes or initiators<br>b) to define safety measures to mitigate plausible dependent failures, if necessary<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>In case of dependent failures and inadvertent interference, ISO 26262 puts emphasis on the avoidance of the common cause failure effects and cascading failures, by design hardening safety measures, and by detecting and controlling such failures through safety mechanisms. A dependent failure analysis (DFA) is performed, to confirm the effectiveness of these safety mechanisms and measures.<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for no violation of TLSR by dependent failures are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D6854_0 | CV2A_DFA_spreadsheet.xlsx, V1-R3 | Dependent Failure Analysis |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-9, Clause 6<br>ISO 26262-9, Clause 7 | Criteria for coexistence of elements<br>Analysis of dependent failures | |

| SC-01.04.03.01 SOLUTION: Avoidance and Control of Dependent Failures |
|---|
| Owner: Alan Zhu, Praveen Jaini, Samy |
| **ARGUMENT** |
| 1. Objective:<br>The analysis of dependent failures aims to identify the single events or single causes that could bypass or invalidate a required independence or freedom from interference between given elements and violate a safety requirement.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>A qualitative Analysis of Dependent Failures (DFA) was performed on the CV2A as planned in the Safety Plan according to the defined processes. The DFA shows that both systematic and random sources for dependent failures were addressed.<br>- For all applicable failure sources of ISO 26262-9 §7.4.4, sufficient measures have been identified and documented in the DFA. |

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- In addition, the DFA considered the typical cascading fault sources and common cause Initiators (CCI) in the context of the usage scenarios of the sensor system and the specific design.
- As the CV2A contains no redundant functionality (single channel system only), no dependencies of redundant functions could be analyzed.
- The CV2A has been analyzed in detail for any dependencies of diagnostic and diagnosed functionality. All identified common elements are monitored by safety mechanisms. Therefore, no single point failure can cause a failure of multiple elements and the judgment of the expert group classified this fault combination as acceptable for the DFA.
- As power supply and other central infrastructure design elements (e.g. clock, reset, DFT, …) can be an important contributor to dependent failures, special focus is put to isolation and other appropriate measures.

The DFA yielded no evidence that the hardware design is not compliant with the requirements on independence. The identified action Items were analyzed and successfully implemented in the software design. All action items were closed.

3. Gaps & Action items:

None.

4. Conclusion:

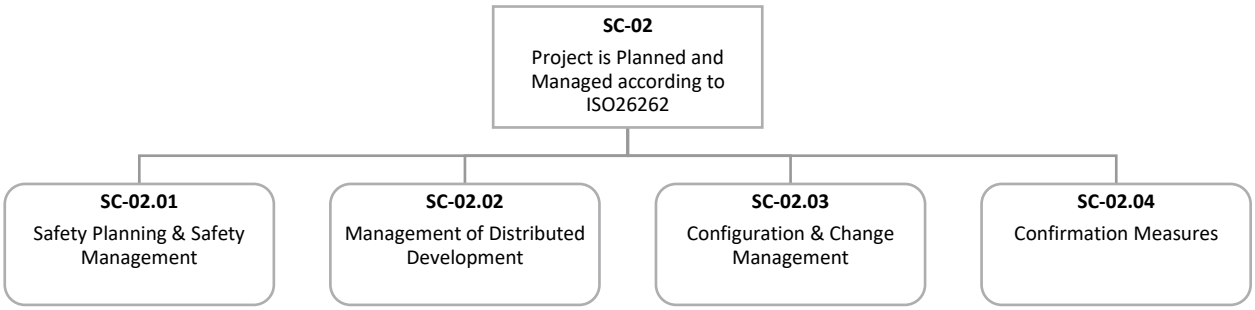
The arguments show that the objectives for avoidance and control of dependent failures are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D6854_0 | cv2a_DFA_spreadsheet.xlsx, V1 R3 | DFA |
| D38600_0 | cv2a_DFA_review_report.docx, v0.5 | DFA verification report |
| D6854_0 | cv2a_DFA_spreadsheet.xlsx, V1 R3, page 3. | DFA Process Description |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-9, Clause 6 | Criteria for coexistence of elements | |
| ISO 26262-9, Clause 7 | Analysis of dependent failures | |
| ISO 26262-5, 7.4.3.5 | Hardware dependent failure analysis | |

## 7.2 Project is Planned and Managed according to ISO 26262 (SC-02)

The following arguments explain the Ambarella safety management, planning and tracking process and how the process meets the applicable ISO 26262-2, -5 and -6 requirements.

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### 7.2.1 Safety Planning & Safety Management (SC-02.01)

| SC-02.01 SOLUTION: Safety Planning & Safety Management | | |
|---|---|---|
| Owner: Asher Shih | | |
| **ARGUMENT** | | |
| 1. Objective:<br>The overall objective of the safety plan is the planning and coordination of the safety activities in the progression of the safety lifecycle, the creation of the safety case, and the execution of the confirmation measures.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>2.1 Argument on Functional Safety Plan:<br>The functional safety plan is prepared based on the *exida* template, "ISO 26262 Functional Safety Management Plan". It specifies the responsibilities to the project management during the safety life cycle is documented, and describes the activities to develop the applicable work products according to ISO 26262:2018. The ISO 26262 life cycle was tailored to the scope of the CV2A chip, which was developed as a HW safety element out of the context (SEooC).<br><br>2.2 Arguments on Functional Safety Management:<br>The safety activities are managed based on the safety plan throughout the safety lifecycle. The status of ISO 26262 work products is monitored and documented in CV2A_WP_doc_mapping_table.xlsx [D37409_0].<br><br>3. Gaps & Action items:<br>None.<br><br>4. Conclusion:<br>The arguments show that the objectives for safety planning and safety management are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D783_2 | CV2A_Safety_Plan.docx, v0.32 | Safety Plan |

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| | | |
|---|---|---|
| D37409_0 | CV2A_WP_doc_mapping_table.xlsx, v0.17 | Status of ISO 26262 work products |
| D783_1 | CV2A_FSMP_RACI_chart.xlsx, v0.15 | RACI Chart |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-2, Clause 5 | Overall safety management | |
| ISO 26262-2, Clause 6 | Project dependent safety management | |
| ISO 26262-2, Clause 7 | Safety management regarding production, operation, service and decommissioning | |
| ISO 26262-4, Clause 5 | General topics for the product development at the system level | |
| ISO 26262-5, Clause 5 | General topics for the product development at the hardware level | |
| ISO26262-8, Clause 10 | Documentation Management | |

### 7.2.2 Management of Distributed Development (SC-02.02)

| **SC-02.02 SOLUTION: Management of Distributed Development** |
|---|
| Owner: Asher Shih, Yu-Ching Su, Argen Huang, Robin Lu |
| **ARGUMENT** |
| 1. Objective:<br>The objectives of this clause are:<br>a) to define the interactions and dependencies between customers and suppliers for development activities<br>b) to describe the allocation of responsibilities<br>c) to identify the work products to be exchanged for distributed developments of the SEooC<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>2.1 Argument on Selection Process:<br>Deep evaluation of foundries was performed from technology, business, design enablement, auto-product support and operation level support.<br><br>2.2 Arguments on DIA:<br>DIA was developed for IP vendor (Samsung), foundry (Samsung), assembly house (ASECL), and testing house (Sigurd). The responsibilities of the work products to be exchanged are specified in DIA, if applicable. Joint tailoring of safety lifecycle was performed with IP vendor (Samsung) and documented in [D32019_0].<br><br>2.3 Arguments on 3rd Party Management:<br>The agreement status of DIA with Samsung, ASECL, and Sigurd were maintained by the person who is responsible for the work item. In addition, the 3rd parity IP integration flow is documented in chapter 10 of Ambarella_Design_Implementation_Flow.docx [D30718_0]. |

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3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for management of distributed development are met.

| EVIDENCE DOCUMENTS | | |
|---|---|---|
| VLSI-1458 | Ambarella Development Interface Agreement, v0.44 | Development interface agreement (DIA) |
| VLSI-16343 | Ambarella Development Interface Agreement – ASECL, v0.3 | |
| VLSI-16271 | Ambarella Development Interface Agreement – Sigurd, v0.03 | |
| D32019_0 | Tailoring_safety_lifecycle_Samsung.docx, v0.51 | Joint tailoring of safety lifecycle with IP vendor |
| D31426_0 | samsung_foundry_memorycompiler_Safety_Plan_Rev3.00_v1.0_customer_delivery_CV2A.pdf, Rev3.00 | Supplier's safety plan |
| D31426_1 | samsung_foundry_otp40kb_cp_ln10lpp_safety_plan_rev3.00_v1.0_customer_delivery_CV2A.pdf, Revision V3.00 | |
| D43254_0 | Report about the evaluation of the ARM Cortex-R52 Processor IP Core, Report-No. 968/FSP 1923.00/20 | Supplier's functional safety assessment report |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-8, Clause 5 | Interfaces within distributed developments | |

### 7.2.3 Configuration & Change Management (SC-02.03)

| SC-02.03 SOLUTION: Configuration & Change Management |
|---|
| Owner: Ray Ko, Asher Shih, Zhen Zhou, Yu-Ching Su, Chan Lee |
| **ARGUMENT** |
| 1. Objective:<br>1.1 The objectives of Configuration Management are to ensure that the work products, the SEooC and the principles and general conditions of their creation can be uniquely identified and reproduced in a controlled manner at any time, and to ensure that the relations and differences between earlier and current versions can be traced.<br>1.2 The objectives of Change Management is to analyze and control changes to safety-related work products, and the SEooC throughout the safety lifecycle. |

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2. Argument for meeting the objective and related ISO 26262 requirements

2.1 Argument on Configuration Management:

Configuration Management plan is split into two phases, pre-silicon and post-silicon. In pre-silicon phase, the version of work products are maintained by either Polarion, SOS, or svn. The details of the configuration management is documented in the Configuration Management Plan [D94_0]. In post-silicon phase, the version of work products are maintained in either svn, git, Document Management System (DMS), or AEC-Q100 server. The configuration management plan for post-silicon phase is document in [3-CM-001].

2.2 Arguments on Change Management Planning:

The change management plan is split into two phases, before and after release for production. Before release for production, the flow to analyze and control changes is documented in the change management plan [D5070_0]. For the changes after release for production, the changes are managed by Procedure for Engineering Change Control [2-R-003].

2.3 Arguments on Impact Analysis:

A template of impact analysis report [D5071_0] is given for the change request initiator to analyze the impact to safety. The analysis result is documented by the change request initiator, and reviewed by the Safety Manager.

3. Gaps & Action items:

None.

4. Conclusion:

The arguments show that the objectives for configuration and change management are met.

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D94_0 | CV2A_CM_Plan.docx, v0.6 | Configuration Management Plan |
| 3-CM-001 | CV2A Configuration Management Plan for Post-silicon, Version 01 | |
| D5070_0 | CV2A_Change_Management_Plan.docx, v0.17 | Change Management Plan |
| 2-R-003 | Procedure for Engineering Change Control, Rev. 09 | |
| - | Reports under doc/safety/Change_Management/Impact_Analysis | Impact analysis and change request plan |
| - | Logged on the corresponding redmine thread. | Change report |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-8, Clause 7 | Configuration management | |
| ISO 26262-8, Clause 8 | Change management | |
| ISO 26262-2, 6.4.3 | Impact analysis at the item level | |

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| | |
|---|---|
| ISO 26262-2, 6.4.4 | Reuse of an existing element |

### 7.2.4 Confirmation Measures (SC-02.04)

| **SC-02.04 SOLUTION: Confirmation Measures** |
|---|
| Owner: Zhen Zhou |
| **ARGUMENT** |
| 1. Objective:<br>A Confirmation Review includes the checking of correctness with respect to formality, contents, adequacy and completeness regarding the requirements of ISO 26262.<br>A Functional Safety Audit includes the evaluation of the implementation of the required processes against the definitions of the activities referenced or specified in the safety plan.<br>A Functional Safety Assessment includes the evaluation of the achieved functional safety of the SEooC. It results in a provision of a recommendation for acceptance, a conditional acceptance or a rejection and includes the work products required by the safety plan, the implementation of the required processes and a review of the implemented safety measures that can be assessed during the SEooC development.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements<br>2.1 Argument on Confirmation Reviews:<br>The confirmation reviews were carried out as planned in the Safety Plan. The confirmation review report summarized the results of the confirmation reviews of the Safety Plan, the Technical Safety Concept, the safety analyses and the completeness of the safety case for the project. The confirmation reviews show the compliance of the provided documents with the related ISO 26262 requirements.<br><br>2.2 Arguments on Functional Safety Audit:<br>The functional safety audit was performed to evaluate the implementation of the process to see whether the activities specified in functional safety plan were executed accordingly.<br><br>2.3 Arguments on Functional Safety Assessment:<br>*exida* performed the functional safety assessment on CV2A. The assessment includes but not limits to:<br>- functional safety planning<br>- overall functional safety management<br>- detailed review of TSC and its implementation, including safety analyses<br>- detailed review of all ISO 26262 work products relevant and applicable to CV2A.<br>- support processes<br>The result of the functional safety assessment will be documented in an official functional safety assessment report at the end of the last assessment phase.<br><br>3. Gaps & Action items:<br>None. |



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|---|
| 4. Conclusion:<br>The arguments show that the objectives for confirmation measures are met. |

| **EVIDENCE DOCUMENTS** | | |
|---|---|---|
| D783_2 | CV2A_Safety_Plan.docx, v0.32 | Functional Safety Plan |
| D39800_0 | cv2a_confirmation_review_report.xlsx, V1-R2 | Confirmation Review Reports |
| D34763_0 | cv2a_functional_safety_audit_report.docx, v0.6 | Functional Safety Audit Reports |
| D43742_0 | CV2A Assessment and Review Comments_v3r2.docx, Version 3, Revision 2 | Functional Safety Assessment Plan and Report |
| D43306_0 | cv2a_release_for_production_report.docx, v1.0 | Release for Production Report |

| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | |
|---|---|
| ISO 26262-2, 6.4.9 | Confirmation measures |
| ISO 26262-2, 6.4.10 | Confirmation reviews |
| ISO 26262-2, 6.4.11 | Functional safety audit |
| ISO 26262-2, 6.4.12 | Functional safety assessment |
| ISO 26262-2, 6.4.13 | Release for production |

### 7.3 Production Meets Product-Specific Requirements and ISO 26262 standard(SC-03)

| **SC-03 CLAIM: Production Meets Product-Specific Requirements and ISO 26262 standard** |
|---|
| Owner: OPS team |
| **ARGUMENT** |
| 1. Objective:<br>Provide the arguments and evidence that production meets production-specific requirements and ISO 26262 requirements.<br><br>2. Argument for meeting the objective and related ISO 26262 requirements:<br>Ambarella's quality management system is ISO 9001 certified. The requirements for production are properly derived and communicated to Samsung Foundry, ASECL, and Sigurd. They are all IATF-16949 and ISO-9001 certified, and hence have proper automotive production process and quality management.<br><br>3. Gaps & Action items: |

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| | | |
|---|---|---|
| None.<br><br>4. Conclusion:<br>The arguments show that the objectives for production meeting product-specific requirements and ISO 26262 are met. | | |
| **EVIDENCE DOCUMENTS** | | |
| D31407_0 | ASE IATF16949.pdf | The IATF 16949 certification of the manufacturing suppliers |
| D31407_2 | Samsung IATF16949_KR.pdf | |
| D31407_3 | Samsung IATF16949_US_Austin.pdf | |
| D31407_4 | Sigurd IATF16949 | |
| VLSI-1458 | Ambarella Development Interface Agreement, v0.44 | The DIA with Samsung |
| VLSI-16343 | Ambarella Development Interface Agreement – ASECL, v0.3 | The DIA with ASECL |
| VLSI-16271 | Ambarella Development Interface Agreement – Sigurd, v0.03 | The DIA with Sigurd |
| **ISO 26262 CLAUSES AND REQUIREMENTS COVERED** | | |
| ISO 26262-7, Clause 5 | Planning for production, operation, service and decommissioning | |
| ISO 26262-7, Clause 6 | Production | |
| ISO 26262-7, Clause 7 | Operation, service and decommissioning | |

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# 8 Terms and Definitions

ISO 26262-1:2018, clause 3.136 defines *Safety Case* as following:

**Safety Case:** Argument that functional safety (3.67) is achieved for items (3.84), or elements (3.41), and satisfied by evidence compiled from work products (3.185) of activities during development.

The following abbreviations are used throughout this document:

| Abbreviation | Description |
|---|---|
| ADAS | Advanced Driver Assistance System |
| AI | Artificial Intelligence |
| AoU | Assumption-of-Use |
| ASIL | Automotive Safety Integrity Level |
| BIST | Built-in Self Test |
| BSD | Blind Spot Detection |
| CCA / CCF | Common Cause Analysis / Common Cause Failure |
| CCI | Common Cause Initiators |
| CON | Constraint |
| CMS | Camera Monitor System |
| CPU | Central Processing Unit |
| CR | Change Request |
| CRC | Cyclic Redundancy Check |
| CV | Computer Vision |
| DC | Diagnostic Coverage |
| DCLS | Dual-Core Lockstep |
| DFA | Dependent Failure Analysis |
| DFMEA | Design Failure Mode and Effect Analysis |
| DFT | Design For Test |
| DIA | Development Interface Agreement |
| DRAM | Dynamic Random Access Memory |
| DSM | Driver State Monitoring |
| DTTI | Diagnostic Test Time Interval |
| E2E | End-to-End |
| ECC | Error Correction Code |
| FIT | Failures In Time |
| FMEA | Failure Mode and Effect Analysis |
| FMEDA | Failure Mode, Effect and Diagnostic Analysis |
| FSM | Functional Safety Manager / Management |
| FuSa | Functional Safety |
| GSN | Goal Structured Notation |
| HDR | High Dynamic Range |
| HSI | Hardware-Software-Interface Control Document |
| HW | Hardware |

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| Abbreviation | Description |
|---|---|
| HWSR | Hardware Safety Requirement |
| IC | Integrated Circuit |
| ID | Identifier |
| IEC | International Electrotechnical Commission |
| IP | Intellectual Property |
| ISO | International Organization for Standardization |
| LED | Light Emitting Diode |
| LFM | Latent Fault Metric |
| PMHF | Probabilistic Metric of random Hardware Faults |
| QM | Quality Manager / Management |
| QMS | Quality Management System |
| RHWF | Random HW Faults |
| SEooC | Safety Element out of Context |
| SC | Safety Case |
| SG | Safety Goal |
| SM | Safety Mechanism |
| SoC | System on Chip |
| SPFM | Single Point Fault Metric |
| STL | Software Test Library |
| SW | Software |
| SWSR | Software Safety Requirement |
| SYS | System |
| TCL | Tool Confidence Level |
| TLSR | Top-Level Safety Requirement |
| TLSS | Top-Level Safe State |
| TSC | Technical Safety Concept |
| TSR | Technical Safety Requirement |
| V&V | Verification and Validation |
| WP | Work Product |

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